

News Article Classification System

DIY Activity: NLP Project Documentation & Presentation Guide

Project Overview

News platforms publish vast quantities of content daily across various domains. To optimize content organization, recommendation engines, and overall user experience, these articles must be automatically categorized and tagged. This project aims to build a robust NLP system capable of classifying articles using both traditional machine learning and modern deep learning methodologies.

Problem Statement

As digital news consumption scales, manual categorization becomes impossible. The core challenge is to develop a system that can:

- [Automatically classify articles into predefined domains.](#)
- [Provide confidence metrics for downstream applications.](#)
- [Generate concise headlines from full-length article context.](#)

Core Objectives

Task 1: Topic Classification (Core)

The primary goal is to classify articles into four major categories:

- Business
- Sports
- Politics
- Technology

Task 2: Confidence Estimation (Extension)

The system should estimate prediction confidence (Low, Medium, or High) to ensure reliability for automated search and recommendation filters.

Task 3: Headline Generation (Advanced)

Utilizing a Seq2Seq architecture, the system will take article text as input and generate a relevant, human-like headline as output.

Dataset Deep Dive

The project utilizes two primary industry-standard datasets to ensure model robustness across varying text lengths and structures.

- **BBC News** (My Preference)
- **AG News**

Technical Workflow

1. Data Preparation

- **Feature Merging:** For AG News, we combine the title and description to provide richer context.
- **Text Cleaning:** Lowercasing, punctuation removal, and whitespace normalization to reduce noise.

2. Feature Engineering

- **TF-IDF:** Providing a baseline statistical representation of word importance.
- **N-grams:** Capturing contextual phrases (e.g., "artificial intelligence").
- **Word Embeddings:** Utilizing dense vectors (Word2Vec/GloVe) for semantic understanding.

3. Modeling Progression

- **Baseline:** Logistic Regression, Nave Byes, D. Tree, XG Boost, etc.. for rapid benchmarking

- **Deep Learning:** Embedding-based models and RNN/LSTM architectures for sequence modeling.
- **Generative:** Seq2Seq (or T5) for the headline generation task.

Performance Evaluation

- **Accuracy:** Overall correct prediction rate.
- **Precision & Recall:** Critical for understanding class-specific errors (e.g., Business vs. Politics overlap).
- **Confusion Matrix:** Visualizing where the model fumbles.

Key Learning Outcomes

- Mastering end-to-end NLP pipelines from messy text to structured predictions.
- Understanding the performance difference between traditional TF-IDF and deep sequence models (LSTM).
- Gaining experience in text generation tasks (Seq2Seq), moving beyond simple classification.
- Evaluating real-world impact on content discoverability and automated summarization.